

A Schoolroom Derivation of the Chandrasekhar Limit

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Aim. We reconstruct the Chandrasekhar mass

$$M_{\text{limit}} = \frac{\omega_3^0 \sqrt{3\pi}}{2} \left(\frac{\hbar c}{G} \right)^{3/2} \frac{1}{(\mu_e m_H)^2}$$

using only ingredients that sit inside the IB Physics and Mathematics syllabus: the de Broglie relation, the Heisenberg uncertainty principle, the Pauli exclusion principle, gravitational self-energy, and the relativistic energy relation. We will recover the entire physical structure of the formula honestly, and we will be candid that the leading dimensionless number, $\omega_3^0 \sqrt{3\pi}/2 \approx 3.10$, is the one piece that genuinely requires the differential equation of stellar structure and so will be quoted rather than derived.

1 Reading the formula before we derive it

Before deriving anything it is worth pausing on the formula itself and sorting its ingredients into the two categories that dimensional analysis insists upon, namely the pure numbers that carry no units at all and the dimensionful constants that do, because once the two are separated the physical meaning of the whole expression becomes transparent. Three of the factors are pure numbers, the Lane–Emden surface constant $\omega_3^0 \approx 2.018$, the geometric factor $\sqrt{3\pi}/2 \approx 1.535$, and the mean molecular weight per electron μ_e , which is a counting ratio and therefore dimensionless; everything that actually carries dimensions is gathered into the fundamental constants $\hbar$, c , G , and the hydrogen mass m_H .

It also helps to see in advance where the exponents of the formula will come from, because the architecture of the result is dictated almost entirely by the geometry of a sphere. Newton’s shell theorem, which is the gravitational content of Gauss’s law, allows a spherically symmetric body to be replaced for every external purpose by a single point mass M concentrated at its centre, so that the field felt at the surface is $g = GM/R^2$, the inverse square being the direct imprint of the $4\pi R^2$ of surface over which that field is spread. The mass is in turn a volume quantity,

$$M = \frac{4}{3}\pi R^3 \rho, \tag{1}$$

growing as the cube of the radius, and these two statements, the volume law R^3 for how much matter is present and the inverse-square surface law R^2 for how its pull thins with distance, are very nearly the whole of the geometry the derivation will call upon. Translated into a count of particles they become the exponents that recur from here onwards, since packing N particles at a fixed spacing fills a volume and so fixes the radius through $R \propto N^{1/3}$. Now picture a single nucleon of mass m_H acting as a test particle at the surface, held there against the pull of the entire mass M beneath it, which the shell theorem lets us treat as a point at the centre; the gravitational binding energy of that one particle is GMm_H/R , and substituting the volume law for M and cancelling the single power of R in the denominator against the three in the numerator gives

$$\frac{GMm_H}{R} = \frac{G\left(\frac{4}{3}\pi R^3 \rho\right) m_H}{R} = \frac{4}{3}\pi G \rho m_H R^2 \propto R^2 \propto N^{2/3}, \tag{2}$$

so the binding energy of the surface particle grows as the surface area R^2 , and therefore as the two-thirds power of the count. It is worth being clear that M here is the whole mass of the star while m_H is the lone particle whose fate we are following, so the expression is one particle's energy in the field of all the others rather than any product of the total mass with itself. The three-halves power that sits on $\hbar c/G$ in the formula, and the two-thirds powers that will surface in the energy balance, are accordingly not algebraic accidents but the volume and surface laws of three-dimensional space expressed in the language of a particle count.

The decisive observation is that the particular grouping $\hbar c/G$ has the dimensions of a mass squared, which we can verify by writing the constants in base SI units,

$$[\hbar] = \text{kg m}^2 \text{s}^{-1}, \quad [c] = \text{m s}^{-1}, \quad [G] = \text{m}^3 \text{kg}^{-1} \text{s}^{-2}, \quad (3)$$

so that

$$\left[\frac{\hbar c}{G} \right] = \frac{\text{kg m}^2 \text{s}^{-1} \cdot \text{m s}^{-1}}{\text{m}^3 \text{kg}^{-1} \text{s}^{-2}} = \text{kg}^2, \quad (4)$$

and hence $(\hbar c/G)^{1/2}$ is a mass. That mass is the Planck mass,

$$m_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}} \approx 2.18 \times 10^{-8} \text{ kg}, \quad (5)$$

the unique mass that can be built from the constants of quantum mechanics, relativity, and gravitation acting together. The Chandrasekhar expression therefore carries the *cube* of the Planck mass, since $(\hbar c/G)^{3/2} = m_{\text{Pl}}^3$ has the dimension of mass cubed, divided by the square of the hydrogen mass, and gathering the pure numbers onto one side lets us rewrite the formula in the wholly equivalent and far more legible form

$$M_{\text{limit}} = \frac{\omega_3^0 \sqrt{3\pi}}{2 \mu_e^2} \frac{m_{\text{Pl}}^3}{m_H^2}. \quad (6)$$

Read this way the formula makes a striking claim, that the heaviest possible white dwarf is fixed by the ratio of the quantum-gravitational mass scale to the atomic mass scale, with the dimensionful anchor $m_{\text{Pl}}^3/m_H^2 \approx 3.68 \times 10^{30} \text{ kg} \approx 1.85 M_\odot$ setting the scale and the leading pure number $\omega_3^0 \sqrt{3\pi}/(2\mu_e^2) \approx 0.77$ for $\mu_e = 2$ trimming it to the familiar value. The dimensional bookkeeping closes correctly, because mass cubed divided by mass squared returns a mass, and the following table makes the separation explicit.

Symbol	Meaning	Carries units?	Value
ω_3^0	Lane–Emden surface gradient	no (pure)	2.018
$\sqrt{3\pi}/2$	geometry and phase-space factor	no (pure)	1.535
μ_e	nucleons per electron	no (pure)	2 for C/O
$\hbar$	reduced Planck constant	yes	$1.0546 \times 10^{-34} \text{ J s}$
c	speed of light	yes	$2.998 \times 10^8 \text{ m s}^{-1}$
G	gravitational constant	yes	$6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$
m_H	mass of hydrogen atom	yes	$1.674 \times 10^{-27} \text{ kg}$

What μ_e counts, and why it equals two

The quantity μ_e deserves a sentence of its own because it is the only place where the chemistry of the star, rather than the fundamental constants of nature, enters the result. It is defined as the number of nucleons present for every free electron, equivalently the mass of the star measured in hydrogen masses divided by the number of electrons available to supply the degeneracy pressure

that holds the star up. For a fully ionised pure element of mass number A and atomic number Z , each nucleus contributes A nucleons to the mass while releasing exactly its Z electrons, so the ratio is simply

$$\mu_e = \frac{A}{Z}. \quad (7)$$

A carbon–oxygen white dwarf is built from carbon-12 and oxygen-16, and both of these are alpha nuclei in which the number of protons equals the number of neutrons, so that the mass number is exactly twice the atomic number; carbon-12 gives $A/Z = 12/6$ and oxygen-16 gives $A/Z = 16/8$, and each of these ratios equals two. Since every nucleus in the star therefore delivers two nucleons of mass for each electron it surrenders, the mixture has $\mu_e = 2$ irrespective of the precise carbon-to-oxygen proportion, which is why the value can be quoted without specifying the composition exactly. The comparison with other compositions is illuminating, because pure ionised hydrogen with $A/Z = 1$ would give $\mu_e = 1$ and, through the μ_e^2 sitting in the denominator of equation (6), a limiting mass fully four times larger, whereas iron-56 with $A/Z = 56/26 \approx 2.15$ would push the limit slightly lower, so the empirical fact that ordinary stellar fuel burns to carbon and oxygen is written directly into the canonical figure of about 1.44 solar masses.¹

2 The competition that decides a star’s fate

A white dwarf is the cooling core left behind when a star like the Sun exhausts its fuel. Nothing is burning inside it any longer, so there is no thermal pressure from fusion to hold it up, and the question of whether it survives or collapses comes down to a single contest between two effects that both grow as the star is squeezed. Gravity pulls every part of the star towards every other part and therefore favours a smaller radius, while the electrons, forbidden by the Pauli principle from crowding into the same quantum state, generate an outward *degeneracy pressure* that resists compression. Our whole derivation consists of writing down the energy associated with each of these effects, noticing how each one scales with the radius of the star, and asking when the balance between them can no longer be maintained.

We model the star as a sphere of radius R containing N electrons, together with the nuclei that supply the bulk of the mass and the positive charge that keeps the whole object neutral. Recalling that μ_e is the number of nucleons present per electron introduced above, the total mass is essentially the nucleon count times the hydrogen mass,

$$M = N \mu_e m_{\text{H}}, \quad (8)$$

which is the relationship that will eventually let us trade a count of electrons for a stellar mass.

¹This figure of $1.44 M_{\odot}$ is almost certainly the one piece of the subject a reader will have met before, and it is frequently the whole of what is remembered, so it is worth setting down plainly what it is and what it is not. It is the boxed formula of this essay evaluated for a carbon–oxygen composition, that is for $\mu_e = 2$, with the exact Lane–Emden coefficient $\omega_3^0 \sqrt{3\pi}/2 \approx 3.10$ restored in place of the order-unity constant that the scaling argument could not itself supply; the same formula stripped of that coefficient yields only the $0.46 M_{\odot}$ of the numerical section, and the passage between those two numbers is the entire content of the calculation. The precise modern value drifts between roughly 1.4 and $1.46 M_{\odot}$ according to the exact constants adopted and to whether one inserts the hydrogen mass or the atomic mass unit, and Chandrasekhar’s own estimate from the early 1930s sat close to this. The number holds only under a particular and idealised set of conditions, namely a non-rotating, non-magnetic, cold and fully degenerate star treated in Newtonian gravity, and every relaxation of those conditions moves it, since a core of iron rather than carbon and oxygen lowers it to about $1.25 M_{\odot}$ through the μ_e^{-2} dependence discussed below, the general-relativistic correction trims it slightly downward, and rotation can support a little more, so the celebrated 1.44 is best read as the value at a clean reference point rather than as a fixed constant of nature.

3 Gravity wants to crush the star

The gravitational potential energy stored in a self-gravitating sphere is negative, because work must be done against gravity to disperse the material to infinity, and on dimensional grounds it must be built from G , the mass M , and the radius R in the combination

$$E_{\text{grav}} \sim -\frac{GM^2}{R}. \quad (9)$$

A uniform sphere gives the precise coefficient $\frac{3}{5}$, and a realistic centrally condensed star gives something larger, but the exact geometric factor is one of the order-unity numbers we are deliberately setting aside; what matters for the argument is that gravity contributes an energy that scales as $-1/R$, growing ever more negative as the star shrinks.

4 Pauli and Heisenberg supply the pressure

The electrons resist this squeezing for a purely quantum reason. Because no two of them may share a quantum state, packing N electrons into a volume $V \sim R^3$ forces them to a number density

$$n = \frac{N}{V} \sim \frac{N}{R^3}, \quad (10)$$

so the typical spacing between neighbouring electrons is $d \sim n^{-1/3} \sim R/N^{1/3}$. Confining an electron to a region of that size carries an unavoidable cost in momentum, since the uncertainty principle $\Delta x \Delta p \gtrsim \hbar$ means that a small $\Delta x \sim d$ forces a large momentum,

$$p \sim \frac{\hbar}{d} \sim \frac{\hbar N^{1/3}}{R}. \quad (11)$$

This is the same physics as the de Broglie relation $p = h/\lambda$ read in reverse: squeezing the available wavelength drives the momentum up. The keen reader who counts states in momentum space carefully (two spin states per cell of volume $\hbar^3$, filled up to a Fermi momentum) finds the sharper result $p \sim \hbar (3\pi^2 n)^{1/3}$, and it is precisely that factor of 3π which surfaces under the square root in the final formula.

5 The relativistic twist that creates a limit

Everything so far would, on its own, produce a stable star rather than a catastrophe, and to see why a *limit* exists we have to ask how energetic these confined electrons become. At modest densities the electrons move slowly and their kinetic energy follows the familiar $p^2/2m_e$, but as the star is compressed the momentum in equation (11) climbs until the electrons are forced to travel at speeds approaching that of light, at which point the relativistic energy relation

$$E^2 = (pc)^2 + (m_e c^2)^2 \xrightarrow{pc \gg m_e c^2} E \approx pc \quad (12)$$

takes over and each electron carries an energy that grows only *linearly* with its momentum. Using equation (11), the energy of a single ultra-relativistic electron is therefore

$$\varepsilon \sim pc \sim \frac{\hbar c N^{1/3}}{R}, \quad (13)$$

and summing over all N electrons gives the total degeneracy energy

$$E_{\text{deg}} \sim N\varepsilon \sim \frac{\hbar c N^{4/3}}{R}. \quad (14)$$

The decisive feature is that this also scales as $1/R$, exactly the same dependence on radius as the gravitational term in equation (9), which is the structural coincidence that makes a sharp mass limit possible.

6 Balancing the two energies

Adding the gravitational energy (9) and the degeneracy energy (14), and writing the mass in terms of the electron count through equation (8), the total energy of the star takes the revealing form

$$E(R) \sim \frac{1}{R} \left[\hbar c N^{4/3} - G \mu_e^2 m_H^2 N^2 \right]. \quad (15)$$

Because both terms share the same $1/R$ factor, the radius cancels out of the question of which effect wins, and the sign of the bracket alone decides the star's fate. When the bracket is positive the degeneracy term dominates, the energy is positive and falls as the star expands, so the star simply puffs outward until the electrons slow back down into the non-relativistic regime where a genuine stable equilibrium waits. When the bracket is negative gravity dominates, the energy is negative and falls without bound as $R \rightarrow 0$, and there is nothing to arrest the collapse; the star runs away towards a neutron star or a black hole.

7 Solving for the limiting mass

The boundary between survival and collapse is the electron count at which the bracket in equation (15) vanishes,

$$\hbar c N^{4/3} = G \mu_e^2 m_H^2 N^2. \quad (16)$$

Dividing through by $N^{4/3}$ isolates the count,

$$\hbar c = G \mu_e^2 m_H^2 N^{2/3} \quad \implies \quad N_{\max} = \left(\frac{\hbar c}{G \mu_e^2 m_H^2} \right)^{3/2}, \quad (17)$$

and substituting this back into the mass relation $M = N \mu_e m_H$ gives

$$\begin{aligned} M_{\text{limit}} &= \mu_e m_H \frac{(\hbar c)^{3/2}}{G^{3/2} (\mu_e^2 m_H^2)^{3/2}} = \mu_e m_H \frac{(\hbar c)^{3/2}}{G^{3/2} \mu_e^3 m_H^3} \\ &= \left[\left(\frac{\hbar c}{G} \right)^{3/2} \frac{1}{(\mu_e m_H)^2} \right]. \end{aligned} \quad (18)$$

This is the printed formula in every respect except for the leading dimensionless constant, and a schoolroom energy argument of this kind can never pin that constant down because we threw away all the order-unity geometric factors along the way. Solving the actual hydrostatic equation of a relativistic degenerate gas (the Lane–Emden equation of index three) replaces our anonymous “ $\sim$ ” with the exact value

$$\frac{\omega_3^0 \sqrt{3\pi}}{2} = \frac{2.01824 \times \sqrt{3\pi}}{2} \approx 3.10, \quad (19)$$

where ω_3^0 is the dimensionless surface gradient of the Lane–Emden solution. The square root of 3π is the fingerprint of the phase-space counting flagged in Section 4, so even the apparently mysterious $\sqrt{3\pi}$ has an honest origin in the same physics we have used throughout.

8 Putting in the numbers

Having already identified the dimensionful anchor $m_{\text{Pl}}^3/m_{\text{H}}^2 \approx 1.85 M_{\odot}$ in equation (6), the arithmetic now amounts to applying the pure-number prefactor to it. Taking a carbon–oxygen white dwarf with $\mu_e = 2$, our bare formula (18) with a coefficient of order one keeps only the $1/\mu_e^2$ and gives

$$M_{\text{limit}} \approx \frac{m_{\text{Pl}}^3}{(\mu_e m_{\text{H}})^2} = \frac{1}{\mu_e^2} \frac{m_{\text{Pl}}^3}{m_{\text{H}}^2} \approx \frac{1}{4} \times 1.85 M_{\odot} \approx 9.2 \times 10^{29} \text{ kg} \approx 0.46 M_{\odot}, \quad (20)$$

which already lands within a factor of three of the truth from order-of-magnitude reasoning alone. Restoring the Lane–Emden coefficient $\omega_3^0 \sqrt{3\pi}/2 \approx 3.10$ multiplies this up to

$$M_{\text{limit}} \approx 3.10 \times 9.2 \times 10^{29} \text{ kg} \approx 2.85 \times 10^{30} \text{ kg} \approx 1.44 M_{\odot}, \quad (21)$$

the celebrated value of about 1.44 solar masses above which no white dwarf can survive.

9 Stellar context: two endpoints and the part played by composition

The same limiting mass presides over two quite different stellar deaths, and it is worth setting them side by side, since they share the formula while differing in almost everything else. The first arises in a genuinely massive star above roughly eight to ten solar masses, which burns its way up the full ladder of nuclear fuels and arranges the products in concentric shells of hydrogen, helium, carbon, neon, oxygen and silicon surrounding a growing core of iron; because iron sits at the peak of the binding-energy curve no further fusion can release energy, so the core merely accumulates until its mass crosses the Chandrasekhar threshold, whereupon electron degeneracy fails and the core collapses within a fraction of a second into a neutron star, or into a black hole if it is heavy enough, hurling the overlying shells outward as a core-collapse supernova. The second endpoint belongs to a carbon–oxygen white dwarf that is not isolated but is drawing matter from a binary companion, so that its mass creeps upward towards the very same limit; here the approach to the threshold drives the central temperature high enough to ignite the carbon almost all at once, and the star is disrupted completely as a Type Ia supernova, leaving no remnant at all rather than a neutron star.

Composition enters the formula only through the mean molecular weight per electron, and the reason the figure of 1.44 solar masses can be quoted so freely without qualification is that μ_e is remarkably stable across the fuels that matter. For a mixture the defining relation is the electron-weighted sum

$$\frac{1}{\mu_e} = \sum_i X_i \frac{Z_i}{A_i}, \quad (22)$$

in which X_i is the mass fraction of species i , and the decisive fact is that every nucleus on the so-called alpha ladder, running from helium-4 through carbon-12, oxygen-16, neon-20, magnesium-24 and silicon-28, contains equal numbers of protons and neutrons and so has $Z/A = 1/2$, returning $\mu_e = 2$ identically. A white dwarf or a stellar core assembled from any combination of these therefore carries $\mu_e = 2$ irrespective of the precise proportions, which is why the limit is indifferent to whether the object is mostly carbon, mostly oxygen, or some blend of the heavier alpha nuclei.

The two places where this constancy breaks down are the instructive ones. Hydrogen is the single common species with $Z/A = 1$ rather than $1/2$, so a hydrogen-rich gas of mass fraction X has $\mu_e = 2/(1 + X)$, sliding towards unity for pure hydrogen; this is precisely why the relevant cores must first be stripped of their hydrogen, because the envelopes shed in a planetary nebula or

expelled in a supernova are exactly the hydrogen-bearing material that would otherwise depress μ_e and inflate the limit. At the opposite extreme the iron-peak nuclei carry a slight neutron excess, so that iron-56 has $Z/A = 26/56$ and hence $\mu_e \approx 2.15$, and because the limiting mass scales as μ_e^{-2} this lowers the effective Chandrasekhar mass of an iron core to

$$M_{\text{limit}} \approx 1.44 \left(\frac{2}{2.15} \right)^2 M_{\odot} \approx 1.25 M_{\odot}, \quad (23)$$

before the further reductions that electron capture and thermal corrections supply in the final approach to collapse. Metallicity in the conventional astronomical sense, meaning the inherited abundance of elements heavier than helium, matters chiefly for how much mass the progenitor loses and therefore for which of the two endpoints it eventually reaches, whereas for the degenerate core itself the only attribute that survives into the formula is μ_e , fixed almost entirely by the absence of hydrogen and by the equal proton and neutron counts of the alpha-chain ash.

What we did and did not earn

The physics of why a maximum mass exists at all is entirely within reach of a strong IB student, because the heart of it is the recognition that the relativistic degeneracy energy and the gravitational energy both scale as $1/R$, so that the radius drops out and a single critical mass is left behind, depending only on the fundamental constants $\hbar$, c and G together with the chemistry encoded in μ_e . The one thing a hand argument cannot supply is the precise numerical coefficient, which belongs to the differential equation of stellar structure and is fairly quoted rather than faked; acknowledging that boundary is itself part of the lesson, since it shows students exactly where the elementary reasoning ends and the machinery of mathematical physics begins.

Addendum: two coupling constants and the shapes of worlds

The cube of the mass ratio that controls the Chandrasekhar limit is the visible face of a quantity that runs through the whole of physics, the gravitational coupling constant, and setting it beside its far more famous electromagnetic cousin is the quickest route to understanding why stars and planets possess the sizes they do. The electromagnetic fine-structure constant measures the strength of the electric interaction between two elementary charges in units fixed by quantum mechanics and relativity,

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137} \approx 7.3 \times 10^{-3}, \quad (24)$$

and its gravitational analogue is assembled in precisely the same fashion from the gravitational attraction between two protons,

$$\alpha_G = \frac{Gm_{\text{H}}^2}{\hbar c} \approx 5.9 \times 10^{-39}. \quad (25)$$

Because both are dimensionless ratios of a force to the same quantum-relativistic yardstick, their quotient is just the ratio of the electric to the gravitational force between two protons,

$$\frac{\alpha}{\alpha_G} = \frac{e^2/4\pi\epsilon_0}{Gm_{\text{H}}^2} \approx 1.2 \times 10^{36}, \quad (26)$$

the celebrated thirty-six orders of magnitude by which electromagnetism overwhelms gravity at the level of single particles, and it is the smallness of α_G that compels astronomical bodies to gather such staggering numbers of atoms before gravity is able to assert itself.

The connection to the cubic ratio is immediate, because $\hbar c/G = m_{\text{Pl}}^2$ tells us that $\alpha_G = (m_{\text{H}}/m_{\text{Pl}})^2$, so the Planck-to-hydrogen mass ratio is nothing other than the inverse square root of the gravitational coupling, $m_{\text{Pl}}/m_{\text{H}} = \alpha_G^{-1/2}$. The Chandrasekhar particle count is therefore

$$N_{\text{max}} = \left(\frac{m_{\text{Pl}}}{m_{\text{H}}} \right)^3 = \alpha_G^{-3/2} \approx 2 \times 10^{57}, \quad (27)$$

and the cube in the formula is exactly the assertion that the number of nucleons a white dwarf can support is the gravitational coupling raised to the power minus three-halves, with the weakness of gravity, registered in the tiny value of α_G , being precisely what makes that count so enormous and the resulting mass of order the Sun's.

The very same machinery answers a question that at first appears unrelated, namely why a body of rock or ice must reach a certain size before its own gravity rounds it into a sphere rather than allowing it to persist as an irregular fragment. A small body keeps its shape because chemical bonds, which are electromagnetic and so governed by α , resist deformation, whereas a large body is reshaped once the gravitational energy gained by an atom sliding towards the centre exceeds the energy of the bonds holding it in place. Estimating the gravitational energy per atom for a body of N atoms packed at the Bohr radius a_0 , so that the radius is $R \sim a_0 N^{1/3}$, gives

$$\frac{E_{\text{grav}}}{\text{atom}} \sim \frac{GMm_{\text{H}}}{R} \sim \frac{Gm_{\text{H}}^2 N^{2/3}}{a_0}, \quad (28)$$

while the binding energy of the bonds is of the order of the Rydberg energy that sets the whole of chemistry,

$$E_{\text{bind}} \sim \alpha^2 m_e c^2 \quad (\text{a few tens of electronvolts}). \quad (29)$$

Setting the two comparable and using the Bohr radius $a_0 = \hbar/(\alpha m_e c)$ to eliminate the atomic scale collapses the entire expression onto the two coupling constants,

$$N^{2/3} \sim \frac{\alpha}{\alpha_G} \quad \implies \quad N \sim \left(\frac{\alpha}{\alpha_G} \right)^{3/2} \approx 10^{54}, \quad (30)$$

so that a cold body crosses over from a chemically governed fragment to a gravitationally governed sphere once it gathers something of the order of 10^{54} atoms, a mass comparable to that of a large planet. The reappearance of the three-halves power is no accident, since it springs from the same geometrical fact established at the outset, that the gravitational energy binding a particle at the surface grows as $N^{2/3}$, so that inverting the relation to find a critical count always returns the exponent $3/2$.

Placing the two results side by side exposes a relationship of real elegance, because the maximum atom count of a planet and the maximum nucleon count of a white dwarf differ only by a factor of the fine-structure constant,

$$\frac{N_{\text{star}}}{N_{\text{planet}}} \sim \frac{\alpha_G^{-3/2}}{(\alpha/\alpha_G)^{3/2}} = \alpha^{-3/2} \approx 137^{3/2} \approx 1.6 \times 10^3, \quad (31)$$

so that the distinction between a body massive enough to ignite as a star and one that remains a cold sphere is written entirely in α , the gravitational coupling having cancelled out completely. The two great structures of the cold and the burning universe are in this way governed by the same pair of pure numbers, the electromagnetic strength that binds atoms into matter and the gravitational strength that binds matter into worlds, and the cubic ratio of the Chandrasekhar formula is simply the most dramatic stage on which the second of them announces how astonishingly feeble it is.

It is worth being candid about the precision of these figures, because the count of 10^{54} marks the scale at which gravity overwhelms chemical binding outright, corresponding to a planet of roughly

Jovian mass, whereas the gentler threshold at which a body merely becomes visibly round sits lower, since partial rounding asks only that gravity exceed the material's yield strength rather than the full bond energy, and that softer criterion, together with the realistic atomic masses of rock that are some tens of times the hydrogen mass, brings the count down towards the range 10^{46} to 10^{50} atoms, the few-hundred-kilometre scale familiar from the larger asteroids and moons. The order-of-magnitude character of these estimates is a feature rather than a flaw, because their purpose is to demonstrate that the sizes of planets and stars are not arbitrary accidents of cosmic history but are fixed, to within the factors of order unity that a fuller theory supplies, by two pure numbers that can be measured on a laboratory bench.

Endnotes

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